# Scoliosis: Exercise and Equipment Guide

Clinical exercise guide · Idiopathic, congenital and neuromuscular scoliosis

## 1. Introduction: understanding the three main categories

Scoliosis is a three-dimensional spinal deformity involving lateral curvature, vertebral rotation and changes to the normal sagittal spinal profile. It is generally identified radiologically when the Cobb angle is at least 10°.

The three main categories classified by cause are:

1. Idiopathic scoliosis

The most common category. The cause is unknown, and it is classified according to age of onset (infantile, juvenile or adolescent). Adolescent idiopathic scoliosis is particularly common during periods of rapid growth.

2. Congenital scoliosis

Develops because of abnormal vertebral formation or segmentation during foetal development. Examples include hemivertebrae and vertebral fusion. The structural abnormality is present from birth, although the curve may become noticeable later.

3. Neuromuscular scoliosis

Develops secondary to neurological or muscular conditions that affect spinal control, muscle strength and balance. Examples include cerebral palsy, muscular dystrophy and spinal muscular atrophy.

Important clinical distinction: exercise does not have the same role in all three categories. Idiopathic scoliosis has the strongest evidence for scoliosis-specific exercise approaches. Congenital scoliosis may involve fixed structural abnormalities that exercise cannot correct. Neuromuscular scoliosis requires an individualised approach based on the underlying condition, functional ability and respiratory health.

The objective of this guide is to understand how equipment can support rehabilitation, improve function, develop muscular capacity and help manage symptoms without making unrealistic claims about correcting spinal deformity.

# 2. Idiopathic scoliosis: exercise and equipment guide

## Main clinical objectives

- Improve postural awareness and active spinal self-correction.
- Develop symmetrical or curve-specific muscular control.
- Improve trunk stability and endurance.
- Improve breathing mechanics, particularly where thoracic rotation affects rib movement.
- Reduce pain and improve physical function.
- Support spinal management during growth.

## Equipment guide

A. Wall bars (stall bars)

Wall Bar Swedish Ladder for Physiotherapy | Rehabilitation - Meddey.com

| Area | Explanation |
|---|---|
| Function | Provide a stable support for balance, stretching, positioning and controlled spinal elongation. |
| Exercise goal | Improve postural awareness, active self-correction and trunk control. |
| What they will not do | They will not mechanically straighten vertebrae or permanently reverse a structural curve. |
| Misconception | Hanging from wall bars automatically stretches the spine into alignment. |
| Safe use | Maintain controlled movement, avoid forceful traction and use a therapist-prescribed correction position. |

Example exercise: supported spinal elongation

1. Stand facing or alongside the wall bars.
2. Hold a comfortable bar position.
3. Lengthen the trunk upwards without forcing the lower back into an exaggerated arch.
4. Apply the curve-specific postural correction taught by a physiotherapist.
5. Maintain normal breathing.

6. Hold for approximately 5–10 seconds initially, repeating 5–8 times if tolerated.

Progression: Begin with supported standing, then develop the ability to maintain the corrected posture without holding the bars.

B. Resistance bands

Resistance Band Row by Brittany D. - Exercise How-to - Skimble

| Area | Explanation |
|---|---|
| Function | Provide adjustable resistance for strengthening the back, shoulders, trunk and hips. |
| Exercise goal | Improve muscular endurance, scapular control and functional strength. |
| What they will not do | They cannot pull the spine into a corrected position or eliminate vertebral rotation. |
| Misconception | Strengthening the weaker-looking side alone will correct the scoliosis. |
| Safe use | Start with light resistance, controlled range of motion and individualised positioning. Avoid uncontrolled twisting or breath-holding. |

Example exercise: seated resistance-band row

1. Sit upright with the band securely anchored.
2. Maintain a comfortable, self-corrected spinal position.
3. Pull the elbows backwards while keeping the shoulders relaxed.
4. Slowly return to the starting position.
5. Start with 1–2 sets of 8–12 repetitions.

Progression: Increase repetitions first, then resistance, while maintaining technique.

C. Swiss ball (stability ball)

Renew Rehab Clinic | Best Physiotherapy, Massage & Acupuncture in Newmarket Toronto

| Area | Explanation |
|---|---|

| | |
|---|---|
| Function | Introduces a controlled unstable surface that challenges balance and trunk muscle activation. |
| Exercise goal | Improve postural reactions, core endurance and coordination. |
| What it will not do | It does not reshape the spine or guarantee symmetrical muscle development. |
| Misconception | Sitting on a Swiss ball all day corrects poor posture and scoliosis. |
| Safe use | Use a correctly sized ball, stable surroundings and supervision where balance is limited. |

Example exercise: seated balance control

1. Sit centrally on the ball with feet shoulder-width apart.
2. Maintain a comfortable neutral or individually corrected posture.
3. Gently transfer weight from side to side.
4. Return to the centre without collapsing through the trunk.
5. Start with 2 sets of 30 seconds.

Progression: Introduce small, controlled arm movements, then more challenging balance tasks.

D. Mirror and postural feedback

Specialized physical therapy helps teens with scoliosis get ahead of the curve | Faculty of Rehabilitation Medicine

| Area | Explanation |
|---|---|
| Function | Provides visual feedback to help patients recognise their spinal and shoulder positioning. |
| Exercise goal | Develop body awareness and the ability to actively self-correct. |
| What it will not do | A mirror cannot diagnose the curve or establish the correct correction direction independently. |

| | |
|---|---|
| Misconception | Making the shoulders look level means the spine is corrected. |
| Safe use | Use alongside an individual assessment and a physiotherapist's instructions. |

Example exercise: Practise a therapist-taught self-correction position in front of the mirror for 5–10 seconds, then relax and repeat.

### Clinical approach: Schroth and PSSE

One of the most relevant evidence-based approaches for idiopathic scoliosis is Schroth therapy.

Physiotherapeutic scoliosis-specific exercises (PSSE) use individually prescribed movements, breathing techniques and postural corrections based on the patient's curve pattern.

Unlike general strengthening, these exercises aim to address the three-dimensional nature of scoliosis.

Clinical point: Do not prescribe the same direction of lateral bending, rotation or asymmetric strengthening to every patient. The correction strategy depends on the individual curve.

## 3. Congenital scoliosis: exercise and equipment guide

### Main clinical objectives

Congenital scoliosis requires a different rehabilitation strategy because the underlying problem is an abnormality in vertebral development rather than simply muscular imbalance.

Exercise is primarily used to:

- Maintain mobility in unaffected spinal segments and joints.
- Improve general muscular strength and physical conditioning.
- Develop balance and coordination.
- Maintain functional independence.
- Reduce secondary discomfort associated with muscular compensation.
- Support rehabilitation before or after surgery, where appropriate.

Important: Exercise cannot correct a malformed vertebra, reverse a hemivertebra or reliably prevent progression of a structurally driven congenital curve. Orthopaedic monitoring is essential, particularly during growth.

## Equipment guide

A. Exercise mat

Physiotherapie und Osteopathie in München | Marianowicz Zentrum

| Area | Explanation |
|---|---|
| Function | Provides a stable, cushioned surface for floor-based mobility and strengthening. |
| Exercise goal | Improve trunk control, hip mobility and general muscular endurance. |
| What it will not do | It will not correct vertebral formation abnormalities or straighten the spine. |
| Misconception | Floor exercises can reverse a congenital spinal deformity. |
| Safe use | Avoid forced spinal rotation, excessive extension and movements that cause pain. Adapt positions to the patient's structural limitations. |

Example exercise: supported pelvic control

1. Lie on the back with knees bent, if comfortable.
2. Keep feet supported on the mat.
3. Gently engage the abdominal muscles.
4. Perform a small, controlled pelvic movement without forcing the lumbar spine.
5. Relax and repeat.

Start with 1 set of 8 repetitions.

Progression: Increase to 2 sets, then introduce controlled limb movements while maintaining trunk stability.

B. Resistance bands

Physiotherapist Westmount - Physiothérapie Westmount ProCare Inc.

| Area | Explanation |
|---|---|
| Function | Allows progressive strengthening of the upper and lower limbs. |

| | |
|---|---|
| Exercise goal | Improve general strength, endurance and functional capacity. |
| What it will not do | It will not change the congenital shape of the vertebrae. |
| Misconception | Uneven muscle strength is always the primary cause of congenital scoliosis. |
| Safe use | Use light resistance initially, maintain controlled movements and avoid painful positions or excessive spinal loading. |

Example exercise: seated resistance-band chest press

1. Sit in a stable chair with the back supported if required.
2. Position the band securely behind the upper back.
3. Hold the ends of the band.
4. Slowly push both arms forwards.
5. Return under control.

Start with 1–2 sets of 8–10 repetitions.

Progression: Increase repetitions and resistance gradually, provided posture and movement quality remain controlled.

C. Stationary exercise bicycle

Physical Medicine and Rehabilitation - Physiatry

| Area | Explanation |
|---|---|
| Function | Provides low-impact cardiovascular exercise. |
| Exercise goal | Improve cardiovascular fitness, lower-limb endurance and general activity tolerance. |
| What it will not do | It will not correct the spinal curve or prevent structural progression. |

| | |
|---|---|
| Misconception | Cycling automatically strengthens the spinal muscles sufficiently to treat scoliosis. |
| Safe use | Adjust the seat and handlebars to avoid uncomfortable spinal positioning. A recumbent bike may be more suitable where trunk support is needed. |

Example exercise: low-intensity cycling

- Begin with 5–10 minutes at a comfortable resistance.
- Maintain a comfortable trunk position.
- Monitor fatigue, discomfort and breathing.
- Gradually build towards 15–20 minutes as tolerated.

Progression: Increase duration before resistance.

D. Physiotherapy balance equipment

Care Physical Therapy | Top-Rated PT in Southfield, MI

| Area | Explanation |
|---|---|
| Function | Challenges proprioception, balance and neuromuscular coordination. |
| Exercise goal | Improve stability during standing, walking and everyday activities. |
| What it will not do | It will not correct a congenital vertebral abnormality. |
| Misconception | Better balance means the spinal curvature is improving. |
| Safe use | Begin on a firm surface, use hand support and introduce unstable surfaces only when appropriate. |

Example exercise: supported single-leg weight transfer

1. Stand near a stable support.
2. Transfer body weight gently between the legs.
3. Maintain a comfortable upright position.
4. Avoid compensatory trunk movements.
5. Repeat 8–10 times.

Progression: Reduce hand support and introduce more challenging balance surfaces when safe.

### Special consideration: post-surgical rehabilitation

Patients who have undergone spinal fusion or hemivertebra resection require a separate rehabilitation programme.

The equipment and exercises must respect:

- Surgical restrictions.
- Healing and fusion timelines.
- Wound recovery.
- Neurological function.
- The surgeon's instructions regarding spinal movement and loading.

General exercise recommendations should never override post-operative restrictions.

## 4. Neuromuscular scoliosis: exercise and equipment guide

Como uma Cadeira de Rodas Adaptada Pode Prevenir a Escoliose?
What Is a Standing Frame and Who Should Use It?
Physio-Logic - Neurological Rehabilitation
11

### Main clinical objectives

Neuromuscular scoliosis develops because an underlying neurological or muscular condition affects the body's ability to maintain normal spinal alignment.

For example:

- Cerebral palsy can involve altered muscle tone, spasticity and impaired postural control.
- Muscular dystrophy can cause progressive muscle weakness.
- Spinal muscular atrophy can affect trunk strength, respiratory function and mobility.

The rehabilitation objectives are therefore different from those for idiopathic scoliosis.

The priority is often to maintain function and quality of life rather than attempting to correct the spinal curvature.

Key objectives include:

- Improve supported sitting and standing.

- Maintain available joint range of motion.
- Improve functional strength where possible.
- Reduce the effects of prolonged immobility.
- Support respiratory function where necessary.
- Improve independence with daily activities.
- Reduce discomfort associated with poor positioning.

The Scoliosis Research Society highlights multidisciplinary care, physiotherapy and wheelchair modifications as important elements of non-operative management.

Scoliosis Research Society+1

## Equipment guide

A. Adjustable wheelchair and postural seating system

2021 M-Series

| Area | Explanation |
|---|---|
| Function | Provides pelvic and trunk support, improving seated positioning and stability. |
| Exercise/rehabilitation goal | Improve sitting balance, comfort, functional use of the arms and participation in daily activities. |
| What it will not do | It will not necessarily stop scoliosis progression or permanently correct the spine. |
| Misconception | A perfectly fitted wheelchair will cure neuromuscular scoliosis. |
| Safe use | Requires individual assessment of pelvic alignment, pressure distribution, skin integrity, respiratory comfort and functional needs. |

Example activity: supported seated reaching

1. Position the individual in their prescribed seating system.
2. Ensure the pelvis and trunk are adequately supported.
3. Place an object within comfortable reaching distance.
4. Encourage controlled reaching with one arm.

5. Return to the starting position.
6. Repeat on the other side, according to functional ability.

Start with 5 repetitions per side.

Progression: Gradually increase reaching distance or introduce functional activities such as picking up objects.

B. Standing frame

R82 Meerkat

| Area | Explanation |
|---|---|
| Function | Supports an individual in an upright standing position when independent standing is difficult. |
| Exercise/rehabilitation goal | Support upright positioning, participation, weight-bearing activity and functional access to the environment. |
| What it will not do | It will not straighten a neuromuscular spinal curve or guarantee prevention of osteoporosis. |
| Misconception | Standing frames reverse scoliosis by stretching the spine vertically. |
| Safe use | Requires appropriate assessment of hip and knee range, bone health, joint stability, skin integrity, orthostatic tolerance and positioning. |

Example exercise: supported standing

1. Transfer the individual safely into the standing frame.
2. Secure the prescribed supports.
3. Check pelvic, trunk, knee and foot positioning.
4. Begin with a short period determined by the rehabilitation team.
5. Monitor comfort, skin, fatigue and circulation.

Progression: Increase standing duration gradually according to clinical tolerance and the individual's goals.

Important: NICE advises against using standing frames solely to prevent low bone mineral density in children with cerebral palsy.

NICE+1

C. Supported treadmill or gait-training equipment

What Is Gait Training - Gait Training Facts and Information | Living Spinal - Living Spinal

| Area | Explanation |
|---|---|
| Function | Provides a controlled environment for practising walking and stepping movements. |
| Exercise/rehabilitation goal | Improve walking ability, coordination, mobility and exercise capacity where appropriate. |
| What it will not do | It will not correct spinal curvature or restore neurological function by itself. |
| Misconception | Repetitive walking automatically reverses muscular weakness or scoliosis. |
| Safe use | Requires appropriate assessment of gait, joint stability, fatigue, bone health and respiratory function. Harness support may be required. |

Example exercise: supported walking

1. Assess whether the individual can safely participate.
2. Adjust treadmill speed to a comfortable level.
3. Use the appropriate level of body-weight support.
4. Practise controlled stepping with therapist assistance.
5. Stop if there is excessive fatigue, pain or deterioration in movement quality.

Progression: Increase walking duration and reduce support only when clinically appropriate.

NICE recognises treadmill gait training as a possible way of improving exercise capacity and mobility in people with neurological conditions.

NICE

D. Respiratory training equipment

AirLife® Incentive Spirometer

| Area | Explanation |
|---|---|
| Function | Certain respiratory devices provide feedback or assistance for breathing exercises. |
| Exercise/rehabilitation goal | Support respiratory function, breathing control and management of respiratory complications where indicated. |
| What it will not do | It will not correct the spinal curve or independently prevent respiratory complications. |
| Misconception | Every patient with scoliosis needs an incentive spirometer. |
| Safe use | Select the device following respiratory assessment. Follow prescribed technique, intensity and monitoring. |

Example exercise: controlled diaphragmatic breathing

1. Position the patient comfortably with appropriate trunk support.
2. Place a hand over the abdomen if this is comfortable.
3. Encourage a slow, controlled breath in.
4. Allow a relaxed exhalation.
5. Repeat for a few comfortable cycles.

Progression: Develop breathing control and coordination with functional activity, under professional supervision.

A respiratory physiotherapist may prescribe additional techniques, such as assisted coughing or airway clearance, when the underlying condition requires them. These are not interchangeable with ordinary breathing exercises.

# 5. Comparison: which equipment is most appropriate for each type?

| Equipment | Idiopathic | Congenital | Neuromuscular |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Wall bars | Useful | Optional | Adapted |
| Resistance bands | Useful | Useful | Adapted |
| Swiss ball | Useful | Optional | Often adapted |
| Exercise mat | Useful | Useful | Adapted |
| Stationary bike | Useful | Useful | As tolerated |
| Mirror | Useful | Optional | Optional |
| Postural wheelchair | Not routine | As needed | Important |
| Standing frame | Not routine | As needed | Important |
| Gait-training equipment | Optional | Optional | As indicated |
| Respiratory equipment | As indicated | As indicated | As indicated |

Useful does not mean essential or suitable for every patient. The selection depends on the person's age, curve pattern, severity, functional ability and clinical assessment.

A clinician must prescribe the actual programme.

### How to build up exercise safely

## Four-stage progression model

1. Stage 1 – Assessment and familiarisation
    - Establish diagnosis and curve characteristics.
    - Identify pain, mobility restrictions and functional limitations.
    - Introduce equipment and basic movements.
    - Establish correct positioning and breathing.
2. Stage 2 – Basic muscular control
    - Introduce low-resistance exercises.
    - Practise controlled movements.

- Develop postural awareness.
- Prioritise technique over repetitions.

3. Stage 3 – Strength and endurance
    - Gradually increase repetitions.
    - Introduce resistance where appropriate.
    - Improve balance and coordination.
    - Integrate exercises into everyday activities.
4. Stage 4 – Functional independence
    - Perform exercises with less assistance where possible.
    - Develop sustainable home exercise routines.
    - Improve participation in daily activities and sport.
    - Reassess clinical outcomes and adjust the programme.

## How to measure whether the exercise programme is working

An important clinical misconception is that improved strength automatically means the scoliosis itself has improved.

These are different outcomes and should be measured separately.

| Assessment | What it measures |
|---|---|
| Cobb angle (radiograph) | Magnitude of spinal curvature |
| Angle of trunk rotation (scoliometer) | Trunk asymmetry and rotational prominence |
| Functional strength testing | Muscular capacity |
| Balance testing | Postural stability |
| Range of motion | Available joint and spinal mobility |
| Six-minute walk test | Functional walking capacity |
| Pain scale | Patient-reported pain |
| SRS-22 questionnaire | Health-related quality of life in scoliosis |

Interpretation: A patient may experience less pain, better balance and greater muscular endurance without a measurable reduction in Cobb angle. This can still represent a successful rehabilitation outcome.

### 7. Safety guidelines and clinical contraindications

Equipment should never be selected simply because it is available in a gym. The patient's diagnosis and functional ability must determine how it is used.

Before exercise:

- Confirm the diagnosis and whether the curve is stable or progressing.
- Establish whether the patient is still growing.
- Identify previous spinal surgery and any restrictions.
- Assess neurological symptoms and muscle strength.
- Consider bone health, joint stability and respiratory function.
- Check whether the patient wears a brace and whether exercises need to be coordinated with brace use.

During exercise:

- Avoid forced spinal correction or aggressive manipulation.
- Do not prescribe generic asymmetric exercises without knowing the curve pattern.
- Avoid uncontrolled twisting and excessive spinal loading.
- Monitor fatigue, breathing, pain and movement quality.
- Ensure wheelchair and standing-frame supports are correctly positioned.
- Progress gradually rather than increasing resistance and duration simultaneously.

When to stop and seek clinical assessment

Stop exercise if the patient experiences new or worsening neurological symptoms, significant or increasing spinal pain, unusual breathlessness, dizziness, faintness or sudden loss of functional ability.

New bladder or bowel dysfunction, saddle numbness or rapidly developing limb weakness requires urgent medical assessment.

## 8. Common misconceptions across all three categories

| Misconception | Clinical reality |
| --- | --- |

| | |
|---|---|
| Exercise can cure every type of scoliosis. | Exercise has different roles depending on the underlying cause. |
| Stronger back muscles automatically straighten the spine. | Strength and spinal alignment are separate outcomes. |
| Stretching the tight side corrects every curve. | Stretching and correction must be individualised. |
| Hanging from a bar decompresses and fixes scoliosis. | Temporary traction sensations do not demonstrate structural correction. |
| A brace and exercise are interchangeable. | Bracing and exercise have different clinical purposes. |
| Neuromuscular scoliosis can be treated with ordinary gym exercises. | Neurological function, fatigue, respiratory health and positioning must be considered. |
| Congenital scoliosis is caused by poor posture. | It results from abnormal vertebral development. |
| Patients with scoliosis should avoid sport. | Most patients can participate in physical activity unless individual restrictions apply. |

The Scoliosis Research Society supports continued physical activity for most people with scoliosis, unless specific medical restrictions apply.

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# 9. Evidence base: what does the research say?

It is important to distinguish between established clinical recommendations and areas where research is still developing.

## Idiopathic scoliosis

Research supports the use of physiotherapeutic scoliosis-specific exercises (PSSE), although the strength of evidence varies.

- The 2016 SOSORT guidelines recommend individualised PSSE, three-dimensional auto-correction, stabilisation of corrected posture and patient education. They also recommend programmes designed by appropriately trained therapists.

PMC

- A 2024 systematic review and meta-analysis found positive effects of PSSE on Cobb angle, trunk rotation and thoracic kyphosis compared with other non-surgical approaches. However, the evidence quality was moderate to low.

PubMed

- A 2024 Cochrane review highlighted uncertainty about how reliably therapeutic exercise prevents progression or avoids more invasive treatment.

PMC

## Congenital scoliosis

- The Scoliosis Research Society explains that congenital scoliosis originates from abnormal vertebral development and can progress during growth. Monitoring and orthopaedic treatment decisions are therefore essential.

Scoliosis Research Society

- The American Academy of Orthopaedic Surgeons explains that physical activity does not increase the risk of curve progression, but treatment depends on the structural abnormality and its severity.

AAOS

## Neuromuscular scoliosis

- The Scoliosis Research Society identifies physiotherapy, wheelchair modification and multidisciplinary management as important non-operative approaches.

Scoliosis Research Society

- NICE guidance supports individualised rehabilitation, positioning and appropriate mobility interventions for people with neurological conditions.

NICE+1

# 10. References and resources

These are the main sources used to develop this guide. They include clinical guidelines, professional organisations and peer-reviewed research.

Clinical guidelines and professional organisations

1. SOSORT (2018). 2016 SOSORT guidelines: Orthopaedic and rehabilitation treatment of idiopathic scoliosis during growth.

Read full clinical guidelines (open access).

2. Scoliosis Research Society. Congenital Scoliosis.

Congenital scoliosis: diagnosis and treatment.

3. Scoliosis Research Society. Neuromuscular Scoliosis.

Neuromuscular scoliosis clinical information.

4. Scoliosis Research Society. Living with Scoliosis.

Physical activity and scoliosis.

5. National Institute for Health and Care Excellence (NICE) (2017, reviewed 2024). Cerebral palsy in under 25s: assessment and management (NG62).

NICE clinical recommendations.

6. American Academy of Orthopaedic Surgeons. Neuromuscular Scoliosis.

Neuromuscular scoliosis and treatment options.

7. American Academy of Orthopaedic Surgeons. Congenital Scoliosis.

Congenital scoliosis: treatment and management.

Peer-reviewed research

8. Baumann, A.N. et al. (2024). The impact of patient scoliosis-specific exercises for adolescent idiopathic scoliosis: A systematic review and meta-analysis of randomized controlled trials with subgroup analysis using observational studies. Spine Deformity.

Read research paper.

9. Effectiveness of Physiotherapeutic Scoliosis-Specific Exercises on 3-Dimensional Spinal Deformities in Patients With Adolescent Idiopathic Scoliosis: A Systematic Review and Meta-analysis (2024).

Read research paper.

10. Cochrane (2024). Therapeutic exercises for idiopathic scoliosis in adolescents.

Read Cochrane systematic review.

11. Wang, Z. et al. (2024). Comparative efficacy of six types of scoliosis-specific exercises on adolescent idiopathic scoliosis: A systematic review and network meta-analysis. BMC Musculoskeletal Disorders, 25, 1070.

Read research paper.